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Redes, sección 10

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Laboratorio 3, parte 2

Packet Tracer

Distribución física general de la red:

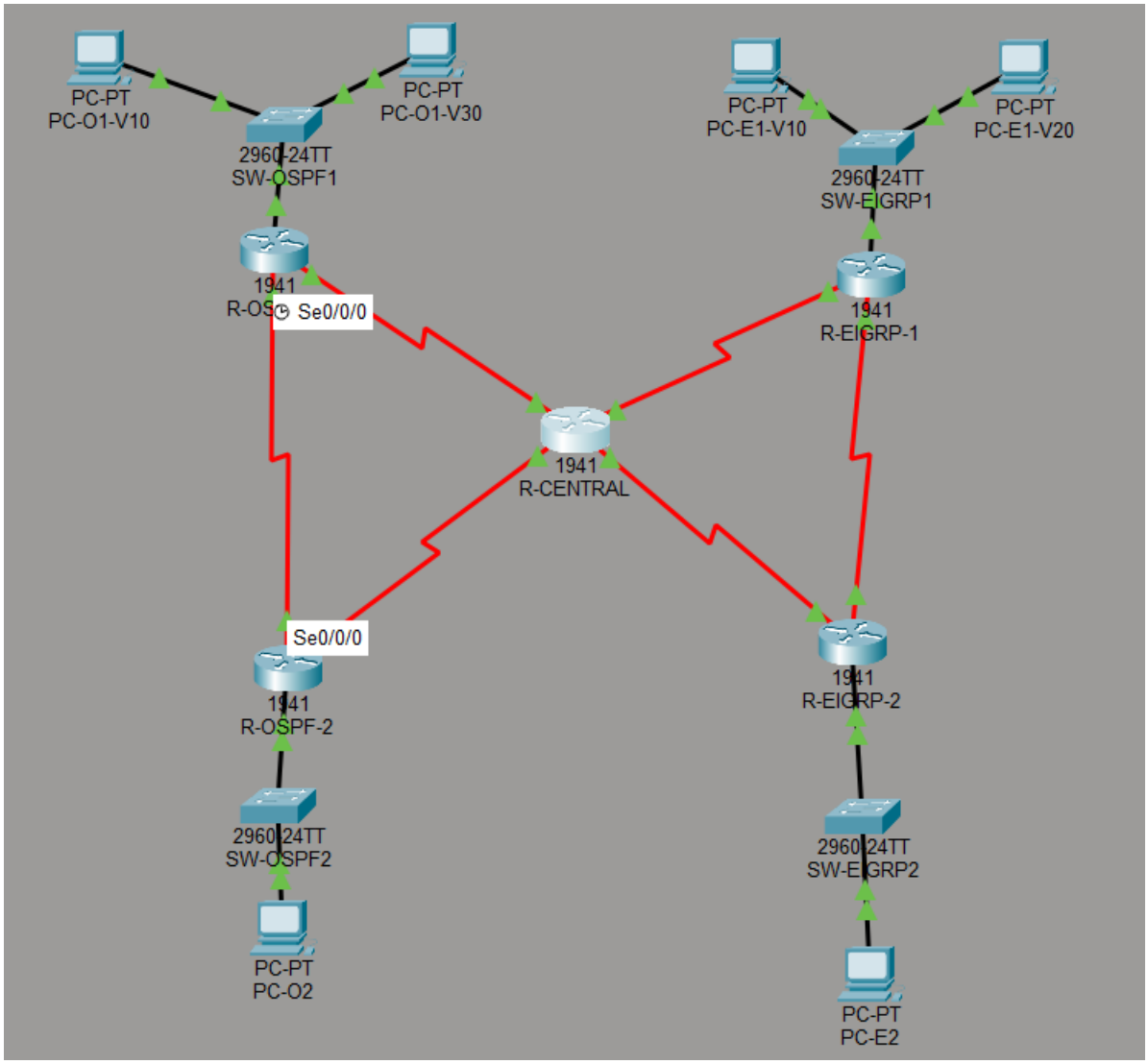


Tabla enrutamiento del router “R-OSPF-1", ejecutando el comando “show ip route”:

```
R-OSPF-1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.1/32 is directly connected, Serial0/0/0
C       10.0.0.4/30 is directly connected, Serial0/0/1
L       10.0.0.5/32 is directly connected, Serial0/0/1
O       10.0.0.8/30 [110/128] via 10.0.0.6, 00:16:55, Serial0/0/1
O E1    10.0.0.16/28 [110/84] via 10.0.0.6, 00:16:55, Serial0/0/1
     172.16.0.0/24 is subnetted, 1 subnets
O E1    172.16.0.0/24 [110/84] via 10.0.0.6, 00:16:55, Serial0/0/1
O    192.168.0.0/24 [110/129] via 10.0.0.2, 00:16:55, Serial0/0/0
                    [110/129] via 10.0.0.6, 00:16:55, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.1.0/26 is directly connected, GigabitEthernet0/0.10
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.1.64/28 is directly connected, GigabitEthernet0/0.30
L       192.168.1.65/32 is directly connected, GigabitEthernet0/0.30

R-OSPF-1#
```

Tabla enrutamiento del router R-OSPF-2, ejecutando el comando “show ip route”:

```
R-OSPF-2>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.2/32 is directly connected, Serial0/0/0
O       10.0.0.4/30 [110/128] via 10.0.0.10, 00:18:53, Serial0/0/1
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.9/32 is directly connected, Serial0/0/1
O E1    10.0.0.16/28 [110/84] via 10.0.0.10, 00:18:53, Serial0/0/1
     172.16.0.0/24 is subnetted, 1 subnets
O E1    172.16.0.0/24 [110/84] via 10.0.0.10, 00:18:53, Serial0/0/1
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/129] via 10.0.0.1, 00:18:43, Serial0/0/0
                       [110/129] via 10.0.0.10, 00:18:43, Serial0/0/1
O       192.168.1.64/28 [110/129] via 10.0.0.1, 00:18:43, Serial0/0/0
                        [110/129] via 10.0.0.10, 00:18:43, Serial0/0/1

R-OSPF-2>
```

Tabla de enrutamiento del router R-CENTRAL, ejecutando el comando “show ip route”:

```
R-CENTRAL>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 11 subnets, 3 masks
O       10.0.0.0/30 [110/192] via 10.0.0.5, 00:19:37, Serial0/0/0
                    [110/192] via 10.0.0.9, 00:19:37, Serial0/0/1
C       10.0.0.4/30 is directly connected, Serial0/0/0
L       10.0.0.6/32 is directly connected, Serial0/0/0
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.10/32 is directly connected, Serial0/0/1
S       10.0.0.16/28 is directly connected, Null0
D       10.0.0.16/30 [90/1536000] via 10.0.0.21, 00:19:57, Serial0/1/0
                     [90/1536000] via 10.0.0.25, 00:19:57, Serial0/1/1
C       10.0.0.20/30 is directly connected, Serial0/1/0
L       10.0.0.22/32 is directly connected, Serial0/1/0
C       10.0.0.24/30 is directly connected, Serial0/1/1
L       10.0.0.26/32 is directly connected, Serial0/1/1
     172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks
S       172.16.0.0/24 is directly connected, Null0
D       172.16.0.0/25 [90/770560] via 10.0.0.25, 00:19:57, Serial0/1/1
D       172.16.0.128/26 [90/770560] via 10.0.0.21, 00:19:57, Serial0/1/0
D       172.16.0.192/26 [90/770560] via 10.0.0.21, 00:19:57, Serial0/1/0
D    192.168.0.0/23 is a summary, 00:19:57, Null0
O    192.168.0.0/24 [110/65] via 10.0.0.9, 00:19:37, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/65] via 10.0.0.5, 00:19:37, Serial0/0/0
O       192.168.1.64/28 [110/65] via 10.0.0.5, 00:19:37, Serial0/0/0

R-CENTRAL>
```

Tabla de enrutamiento del router R-EIGRP-1, ejecutando el comando “show ip route”:

```
R-EIGRP-1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/2585600] via 10.0.0.22, 00:20:54, Serial0/0/1
D EX    10.0.0.4/30 [170/2585600] via 10.0.0.22, 00:21:14, Serial0/0/1
D EX    10.0.0.8/30 [170/2585600] via 10.0.0.22, 00:21:14, Serial0/0/1
C       10.0.0.16/30 is directly connected, Serial0/0/0
L       10.0.0.17/32 is directly connected, Serial0/0/0
C       10.0.0.20/30 is directly connected, Serial0/0/1
L       10.0.0.21/32 is directly connected, Serial0/0/1
D       10.0.0.24/30 [90/3072000] via 10.0.0.18, 00:21:14, Serial0/0/0
                     [90/3072000] via 10.0.0.22, 00:21:14, Serial0/0/1
     172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
D       172.16.0.0/25 [90/1026560] via 10.0.0.18, 00:21:14, Serial0/0/0
                      [90/3074560] via 10.0.0.22, 00:21:14, Serial0/0/1
C       172.16.0.128/26 is directly connected, GigabitEthernet0/0.10
L       172.16.0.129/32 is directly connected, GigabitEthernet0/0.10
C       172.16.0.192/26 is directly connected, GigabitEthernet0/0.20
L       172.16.0.193/32 is directly connected, GigabitEthernet0/0.20
D    192.168.0.0/23 [90/3072000] via 10.0.0.22, 00:21:14, Serial0/0/1

R-EIGRP-1>
```

Tabla de enrutamiento del router R-EIGRP-2, ejecutando el comando “show ip route”:

```
R-EIGRP-2>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/2585600] via 10.0.0.26, 00:21:34, Serial0/0/1
D EX    10.0.0.4/30 [170/2585600] via 10.0.0.26, 00:21:54, Serial0/0/1
D EX    10.0.0.8/30 [170/2585600] via 10.0.0.26, 00:21:54, Serial0/0/1
C       10.0.0.16/30 is directly connected, Serial0/0/0
L       10.0.0.18/32 is directly connected, Serial0/0/0
D       10.0.0.20/30 [90/3072000] via 10.0.0.17, 00:21:54, Serial0/0/0
                     [90/3072000] via 10.0.0.26, 00:21:54, Serial0/0/1
C       10.0.0.24/30 is directly connected, Serial0/0/1
L       10.0.0.25/32 is directly connected, Serial0/0/1
     172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks
C       172.16.0.0/25 is directly connected, GigabitEthernet0/0
L       172.16.0.1/32 is directly connected, GigabitEthernet0/0
D       172.16.0.128/26 [90/1026560] via 10.0.0.17, 00:21:54, Serial0/0/0
                        [90/3074560] via 10.0.0.26, 00:21:54, Serial0/0/1
D       172.16.0.192/26 [90/1026560] via 10.0.0.17, 00:21:54, Serial0/0/0
                        [90/3074560] via 10.0.0.26, 00:21:54, Serial0/0/1
D    192.168.0.0/23 [90/3072000] via 10.0.0.26, 00:21:54, Serial0/0/1

R-EIGRP-2>
```

Demostración de balanceo del protocolo OSPF, bajo el router R-OSPF-1, ejecutando el comando “show ip ospf interface Serial0/0/n”:

```
R-OSPF-1#show ip ospf interface Serial0/0/0

Serial0/0/0 is up, line protocol is up
  Internet address is 10.0.0.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 128
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:02
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
  Suppress hello for 0 neighbor(s)
R-OSPF-1#
```

```
R-OSPF-1#show ip ospf interface Serial0/0/1

Serial0/0/1 is up, line protocol is up
  Internet address is 10.0.0.5/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:03
  Index 4/4, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 5.5.5.5
  Suppress hello for 0 neighbor(s)
R-OSPF-1#
```

Test de balanceo del protocolo OSPF, bajo el router R-OSPF-1, ejecutando el comando "show ip route 192.168.0.0"

```
R-OSPF-1#show ip route 192.168.0.0
Routing entry for 192.168.0.0/24
Known via "ospf 1", distance 110, metric 129, type intra area
  Last update from 10.0.0.2 on Serial0/0/0, 00:24:26 ago
  Routing Descriptor Blocks:
  * 10.0.0.2, from 2.2.2.2, 00:24:26 ago, via Serial0/0/0
      Route metric is 129, traffic share count is 1
    10.0.0.6, from 2.2.2.2, 00:24:26 ago, via Serial0/0/1
      Route metric is 129, traffic share count is 1

R-OSPF-1#
```

Demostración de balanceo del protocolo EIGRP, bajo el router R-EIGRP-1, ejecutando el comando “show ip eigrp topology”

```
R-EIGRP-1>show ip eigrp topology
IP-EIGRP Topology Table for AS 100/ID(172.16.0.193)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - Reply status

P 10.0.0.0/30, 1 successors, FD is 2585600
         via Rstatic (2585600/281600)
P 10.0.0.4/30, 1 successors, FD is 2585600
         via Rstatic (2585600/281600)
P 10.0.0.8/30, 1 successors, FD is 2585600
         via Rstatic (2585600/281600)
P 10.0.0.16/30, 1 successors, FD is 1024000
         via Connected, Serial0/0/0
P 10.0.0.20/30, 1 successors, FD is 2560000
         via Connected, Serial0/0/1
P 10.0.0.24/30, 2 successors, FD is 3072000
         via 10.0.0.18 (3072000/2560000), Serial0/0/0
         via 10.0.0.22 (3072000/768000), Serial0/0/1
P 172.16.0.0/25, 2 successors, FD is 1026560
         via 10.0.0.18 (1026560/28160), Serial0/0/0
         via 10.0.0.22 (3074560/770560), Serial0/0/1
P 172.16.0.128/26, 1 successors, FD is 28160
         via Connected, GigabitEthernet0/0.10
P 172.16.0.192/26, 1 successors, FD is 28160
         via Connected, GigabitEthernet0/0.20
P 192.168.0.0/23, 1 successors, FD is 3072000
         via 10.0.0.22 (3072000/2169856), Serial0/0/1
R-EIGRP-1>
```

Demostración del balanceo del protocolo EIGRP, bajo el router R-EIGRP-1, ejecutando el comando “show ip eigrp topology all-links”

```
R-EIGRP-1>show ip eigrp topology all-links
IP-EIGRP Topology Table for AS 100/ID(172.16.0.193)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - Reply status

P 10.0.0.0/30, 1 successors, FD is 2585600
         via 10.0.0.22 (2585600/281600), Serial0/0/1
P 10.0.0.4/30, 1 successors, FD is 2585600
         via 10.0.0.22 (2585600/281600), Serial0/0/1
P 10.0.0.8/30, 1 successors, FD is 2585600
         via 10.0.0.22 (2585600/281600), Serial0/0/1
P 10.0.0.16/30, 1 successors, FD is 1024000
         via Connected, Serial0/0/0
P 10.0.0.20/30, 1 successors, FD is 2560000
         via Connected, Serial0/0/1
P 10.0.0.24/30, 2 successors, FD is 3072000
         via 10.0.0.18 (3072000/2560000), Serial0/0/0
         via 10.0.0.22 (3072000/768000), Serial0/0/1
P 172.16.0.0/25, 2 successors, FD is 1026560
         via 10.0.0.18 (1026560/28160), Serial0/0/0
         via 10.0.0.22 (3074560/770560), Serial0/0/1
P 172.16.0.128/26, 1 successors, FD is 28160
         via Connected, GigabitEthernet0/0.10
P 172.16.0.192/26, 1 successors, FD is 28160
         via Connected, GigabitEthernet0/0.20
P 192.168.0.0/23, 1 successors, FD is 3072000
         via 10.0.0.22 (3072000/2169856), Serial0/0/1
R-EIGRP-1>
```

Demostración de Variance, bajo el router R-EIGRP-1, ejecutando el comando “show ip protocols”

```
R-EIGRP-1>show ip protocols

Routing Protocol is "eigrp  100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 100
  EIGRP-IPv4 Protocol for AS(100)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
    Router-ID: 10.0.0.17
    Topology : 0 (base)
      Active Timer: 3 min
      Distance: internal 90 external 170
      Maximum path: 4
      Maximum hopcount 100
      Maximum metric variance 3

  Automatic Summarization: disabled
  Automatic address summarization:
  Maximum path: 4
  Routing for Networks:
     10.0.0.16/30
     10.0.0.20/30
     172.16.0.128/26
     172.16.0.192/26
  Passive Interface(s):
    GigabitEthernet0/0.10
    GigabitEthernet0/0.20
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.0.18       90            88874063
    10.0.0.22       90            88874066
  Distance: internal 90 external 170

R-EIGRP-1>
```

Demostración de Variance 2, bajo el router R-EIGRP-2, ejecutando el comando “show ip protocols”

```
R-EIGRP-2>show ip protocols

Routing Protocol is "eigrp  100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 100
  EIGRP-IPv4 Protocol for AS(100)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
    Router-ID: 10.0.0.18
    Topology : 0 (base)
      Active Timer: 3 min
      Distance: internal 90 external 170
      Maximum path: 4
      Maximum hopcount 100
      Maximum metric variance 3

  Automatic Summarization: disabled
  Automatic address summarization:
  Maximum path: 4
  Routing for Networks:
     10.0.0.16/30
     10.0.0.24/30
     172.16.0.0/25
  Passive Interface(s):
    GigabitEthernet0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.0.17       90            88692094
    10.0.0.26       90            88692099
  Distance: internal 90 external 170

R-EIGRP-2>
```

Demostración de redistribution mutua, bajo el router R-CENTRAL, ejecutando el comando “show ip protocols”

```
R-CENTRAL>show ip protocols

Routing Protocol is "eigrp  100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 100, ospf 1
  EIGRP-IPv4 Protocol for AS(100)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
    Router-ID: 10.0.0.6
    Topology : 0 (base)
      Active Timer: 3 min
      Distance: internal 90 external 170
      Maximum path: 4
      Maximum hopcount 100
      Maximum metric variance 1

  Automatic Summarization: disabled
  Automatic address summarization:
    192.168.0.0/23 for Serial0/1/0, Serial0/1/1
      Summarizing with metric 2169856
  Maximum path: 4
  Routing for Networks:
     10.0.0.20/30
     10.0.0.24/30
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.0.21       90            87600616
    10.0.0.25       90            87600617
  Distance: internal 90 external 170


Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 5.5.5.5
  It is an autonomous system boundary router
  Redistributing External Routes from,
```

```
  Redistributing External Routes from,
    static
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.4 0.0.0.3 area 0
    10.0.0.8 0.0.0.3 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:01:29
    2.2.2.2              110      00:01:30
    5.5.5.5              110      00:01:28
  Distance: (default is 110)

R-CENTRAL>
```

Redistribución y semilla, bajo el router R-CENTRAL, ejecutando el comando “show running-config | section router”

```
R-CENTRAL#show running-config | section router
router eigrp 100
 redistribute ospf 1 metric 10000 100 255 1 1500
 network 10.0.0.20 0.0.0.3
 network 10.0.0.24 0.0.0.3
router ospf 1
 router-id 5.5.5.5
 log-adjacency-changes
 redistribute static metric-type 1 subnets
 network 10.0.0.4 0.0.0.3 area 0
 network 10.0.0.8 0.0.0.3 area 0
R-CENTRAL#
```

Prueba total de la conexión, ejecutando desde la computadora PC-O1-V1O el comando “ping 172.16.0.10” (otra PC del otro lado de la arquitectura).

```
C:\>ping 172.16.0.10

Pinging 172.16.0.10 with 32 bytes of data:

Reply from 172.16.0.10: bytes=32 time=7ms TTL=125
Reply from 172.16.0.10: bytes=32 time=6ms TTL=125
Reply from 172.16.0.10: bytes=32 time=9ms TTL=125
Reply from 172.16.0.10: bytes=32 time=17ms TTL=125

Ping statistics for 172.16.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 17ms, Average = 9ms

C:\>
```

Prueba total de la conexión, ejecutando desde la computadora PC-O1-V10 el comando “tracert 172.16.0.10”

```
C:\>tracert 172.16.0.10

Tracing route to 172.16.0.10 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      192.168.1.1
  2    2 ms      4 ms      0 ms      10.0.0.6
  3    4 ms      8 ms      1 ms      10.0.0.25
  4    0 ms      0 ms      3 ms      172.16.0.10

Trace complete.

C:\>
```

Prueba de vecinidad del protocolo OSPF, ejecutado bajo los routers: R-OSPF-1, R-OSPF-2, R-CENTRAL; todos ejecutando el comando: “show ip ospf neighbor”

```
R-OSPF-1>show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           0   FULL/  -        00:00:37    10.0.0.2        Serial0/0/0
5.5.5.5           0   FULL/  -        00:00:37    10.0.0.6        Serial0/0/1
R-OSPF-1>
```

```
R-OSPF-2>show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:36    10.0.0.1        Serial0/0/0
5.5.5.5           0   FULL/  -        00:00:37    10.0.0.10       Serial0/0/1
R-OSPF-2>
```

```
R-CENTRAL#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:30    10.0.0.5        Serial0/0/0
2.2.2.2           0   FULL/  -        00:00:30    10.0.0.9        Serial0/0/1
R-CENTRAL#
```

Demostración de las interfaces pasivas del protocolo OSPF, bajo el router R-OSPF-1, ejecutando el comando “show ip protocols”

```
R-OSPF-1>show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.0 0.0.0.3 area 0
    10.0.0.4 0.0.0.3 area 0
    192.168.1.0 0.0.0.63 area 0
    192.168.1.64 0.0.0.15 area 0
  Passive Interface(s):
    GigabitEthernet0/0.10
    GigabitEthernet0/0.30
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:10:19
    2.2.2.2              110      00:10:20
    5.5.5.5              110      00:10:19
  Distance: (default is 110)

R-OSPF-1>
```

Demostración de las interfaces OSPF pasivas, bajo el router R-OSPF-2, ejecutando el comando “show ip protocols”

```
R-OSPF-2>show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 2.2.2.2
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.0 0.0.0.3 area 0
    10.0.0.8 0.0.0.3 area 0
    192.168.0.0 0.0.0.255 area 0
  Passive Interface(s):
    GigabitEthernet0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:11:02
    2.2.2.2              110      00:11:03
    5.5.5.5              110      00:11:02
  Distance: (default is 110)

R-OSPF-2>
```

Demostración de vecinidad del protocolo y AS100, routers R-EIGRP-1, R-EIGRP-2, R-CENTRAL, “

```
R-EIGRP-1>show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address         Interface       Hold Uptime    SRTT   RTO   Q   Seq
                                    (sec)          (ms)       Cnt Num
0   10.0.0.18       Se0/0/0         13   00:43:21  40     1000  0   58
1   10.0.0.22       Se0/0/1         12   00:43:21  40     1000  0   212

R-EIGRP-1>
```

”

```
R-EIGRP-2>show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address         Interface       Hold Uptime    SRTT   RTO   Q   Seq
                                    (sec)          (ms)       Cnt Num
0   10.0.0.17       Se0/0/0         14   00:43:41  40     1000  0   42
1   10.0.0.26       Se0/0/1         13   00:43:41  40     1000  0   212

R-EIGRP-2>
```

```
R-CENTRAL#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address         Interface       Hold Uptime    SRTT   RTO   Q   Seq
                                    (sec)          (ms)       Cnt Num
0   10.0.0.21       Se0/1/0         14   00:43:56  40     1000  0   43
1   10.0.0.25       Se0/1/1         12   00:43:56  40     1000  0   59

R-CENTRAL#
```

Interfaces EIGRP pasivas, bajo routers R-EIGRP-1 y R-EIGRP-2, ejecutándose el comando: “show ip protocols”

```
R-EIGRP-1>show ip protocols

Routing Protocol is "eigrp  100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 100
  EIGRP-IPv4 Protocol for AS(100)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
    Router-ID: 10.0.0.17
    Topology : 0 (base)
      Active Timer: 3 min
      Distance: internal 90 external 170
      Maximum path: 4
      Maximum hopcount 100
      Maximum metric variance 3

  Automatic Summarization: disabled
  Automatic address summarization:
  Maximum path: 4
  Routing for Networks:
     10.0.0.16/30
     10.0.0.20/30
     172.16.0.128/26
     172.16.0.192/26
  Passive Interface(s):
    GigabitEthernet0/0.10
    GigabitEthernet0/0.20
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.0.18       90            88874063
    10.0.0.22       90            88874066
  Distance: internal 90 external 170

R-EIGRP-1>
```

```
R-EIGRP-2>show ip protocols

Routing Protocol is "eigrp  100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 100
  EIGRP-IPv4 Protocol for AS(100)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
    Router-ID: 10.0.0.18
    Topology : 0 (base)
      Active Timer: 3 min
      Distance: internal 90 external 170
      Maximum path: 4
      Maximum hopcount 100
      Maximum metric variance 3

  Automatic Summarization: disabled
  Automatic address summarization:
  Maximum path: 4
  Routing for Networks:
     10.0.0.16/30
     10.0.0.24/30
     172.16.0.0/25
  Passive Interface(s):
    GigabitEthernet0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.0.17       90            88692094
    10.0.0.26       90            88692099
  Distance: internal 90 external 170

R-EIGRP-2>
```

Demostración de ruteo entre VLANs, bajo los routers R-OSPF-1 y R-EIGRP-1, ejecutando el comando “show ip interface brief”

```
R-OSPF-1>show ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.1.1     YES manual up                    up
GigabitEthernet0/0.30  192.168.1.65    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.0.0.1        YES manual up                    up
Serial0/0/1            10.0.0.5        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R-OSPF-1>
R-OSPF-1>
```

```
R-EIGRP-1>show ip inter br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  172.16.0.129    YES manual up                    up
GigabitEthernet0/0.20  172.16.0.193    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.0.0.17       YES manual up                    up
Serial0/0/1            10.0.0.21       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R-EIGRP-1>
R-EIGRP-1>
```

Demostración de VLANs en modo trunk, bajo los switches SW-OSPF1 y SW-EIGRP1, ejecutando los comandos: “show vlan brief” y “show interfaces trunk”

```
SW-OSPF1>show vlan br

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                 Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                 Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                 Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                 Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                 Fa0/24, Gig0/1, Gig0/2
10   DATOS                             active    Fa0/2
30   ADMIN                             active    Fa0/3
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
SW-OSPF1>
```

```
SW-OSPF1>show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,30

SW-OSPF1>
```

```
SW-EIGRP1>show vlan br

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                 Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                 Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                 Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                 Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                 Fa0/24, Gig0/1, Gig0/2
10   DATOS                             active    Fa0/2
20   VOZ                               active    Fa0/3
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
SW-EIGRP1>
```

```
SW-EIGRP1>show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20

SW-EIGRP1>
```